

Tools that Ensure Web and Smartphone Apps are Safe and Secure

Penetration testing is used to determine the vulnerabilities of Web and smartphone apps. It is also called pen testing or ethical hacking, and can be carried out manually or automatically.



With the increasing dependence on online platforms, the issues and concerns of security and privacy merit attention. A number of cases associated with data breaches, hacking, data leakages, server cracking get reported every day in the global news. This creates a sense of distrust among the public towards the use of online platforms, whether via Web applications or smartphone apps. There is a need to create awareness about the diverse aspects of security and privacy, while using online platforms.

These hacking and data breach attempts actually cost organisations a lot of money, as can be seen in Figure 1 from the reports of the global data platform, Statista.

Free and open source tools for security

There is a need to secure Web and smartphone applications against malicious hackers and data breach attempts. There are a number of tools and frameworks that are widely used for penetration testing and digital forensics. These tools, when used by application developers, prevent data theft.

Vega (<https://subgraph.com/vega/>) is one of the powerful free and open source tools for penetration testing. It is used as a Web security scanner and is an effective platform for Cross Site Scripting (XSS) implementations. In addition, the test associated with SQL injection and information disclosure can be evaluated. Vega is a GUI based tool written in Java and is cross-platform. It is developed by Subgraph and is available for cyber forensic and Web penetration analytics.

The Vega scanner presents the vulnerabilities, with the associated impact classified as either High, Medium or Low. This predicts the performance of the Web application and indicates possible security breach points.

Penetration testing and auditing Android based smartphone apps

Today, most of the Web services and applications are deployed on smartphone platforms including Android, iPhone, Blackberry, KaiOS, Symbian, Java and many others. The software applications launched for mobile platforms need to be tested rigorously against assorted assaults and vulnerability points.